

## Phonetics and Phonology (ENG507)

### ✚ Why Studying Phonetics and Phonology?

- Improved pronunciation: Knowing the correct production and sound of English vowels, diphthongs, and consonants helps learners achieve accurate pronunciation.
- Enhanced language teaching: Teachers can better instruct students on English pronunciation and decoding skills.
- Language learning: Understanding phonetics and phonology aids in learning English and other languages.
- Speech therapy: Studying phonetics and phonology helps speech therapists diagnose and treat speech disorders.
- Language development: Researchers can understand how children acquire language and how language changes over time.

✚ **Focus Language – English:** English is a widely spoken language with a complex sound system, making it an ideal focus for phonetics and phonology studies.

### ✚ Introduction to Vowels and Consonants

- **Vowels:** Sounds made without blocking the airflow in the mouth (e.g., /i/ in "bit").
- **Consonants:** Sounds made by blocking the airflow (e.g., /p/ in "pat").

### ✚ Introduction to English Vowels

English has monophthong vowels (single, pure sounds) and diphthongs (gliding sounds).

- **Monophthong vowels:** /i/ (bit), /e/ (bet), /a/ (bat), /o/ (bot), /u/ (but)
- **Diphthongs:** /ai/ (buy), /au/ (how), /oi/ (boy)

### ✚ Introduction to English Consonants

English consonants include stops, fricatives, nasals, liquids, and approximants.

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- **Stops:** /p/ (pat), /t/ (tot), /k/ (cat)
- **Fricatives:** /s/ (sat), /z/ (zat), /f/ (fat)
- **Nasals:** /m/ (mat), /n/ (nat), /ŋ/ (sing)
- **Liquids:** /l/ (lat), /r/ (rat)
- **Approximants:** /w/ (wat), /j/ (yes)

✚ **IPA Transcription of English Sounds:** The International Phonetic Alphabet (IPA) represents English sounds with unique symbols, ensuring accurate representation and transcription.

✚ **Introduction to Phonology:** Phonology studies the sound system of languages, examining sound distribution, patterns, and rules.

✚ **Introduction to Phonetics:** Phonetics examines the physical properties of speech sounds, including production, acoustic properties, and perception.

## ✚ **Phonetics vs. Phonology**

- **Phonetics:** Focuses on the physical properties of speech sounds.
- **Phonology:** Examines the sound system and patterns of languages.

## ✚ **Types of Phonetic Studies**

- **Articulatory phonetics:** Studies how sounds are produced by the mouth and vocal cords.
- **Acoustic phonetics:** Analyzes the physical properties of speech sounds, like frequency and amplitude.
- **Auditory phonetics:** Examines how listeners perceive speech sounds.
- **Experimental phonetics and phonology:** Employs scientific methods to investigate phonetic and phonological phenomena.
- **Generative phonology:** A theoretical framework that posits rules and constraints governing sound patterns in languages.

## ✚ **Articulatory Phonetics-Speech Production**

Articulatory phonetics investigates how speech sounds are produced by the articulators (lips, tongue, teeth, etc.).

- **Place of articulation:** Where the sound is made (e.g., bilabial /p/).
- **Manner of articulation:** How the sound is made (e.g., stop /p/).
- **Voicing:** Whether the sound is voiced (e.g., /z/) or voiceless (e.g., /s/).

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- ✚ **Sound Waves:** Sound waves are the physical basis of speech. They are created by the vibration of the vocal cords and travel through the air as pressure waves.
- ✚ **The Oro-Nasal Process:** The oro-nasal process refers to the coordination of the oral and nasal cavities in speech production. Sounds can be made with either an oral (mouth) or nasal (nose) articulation, or a combination of both.
- ✚ **Articulatory Gestures:** Articulatory gestures are the movements and positions of the articulators (lips, tongue, teeth, etc.) that produce speech sounds.
- ✚ **Manners of Articulation:** Manners of articulation refer to the way in which the articulators interact to produce sounds.
- ✚ **Stop:** Oral and Nasal
  - Oral stops: /p/, /t/, /k/ (made by blocking the airflow with the lips or tongue)
  - Nasal stops: /m/, /n/, /ŋ/ (made by allowing airflow through the nose)
- ✚ **Fricative:** Fricatives are made by directing airflow through a narrow channel, causing friction: /s/, /z/, /f/, /v/
- ✚ **Approximants:** Approximants are made by directing airflow through a relatively wide channel, with little friction: /w/, /j/, /r/
- ✚ **Trill, Tap and Flap**
  - **Trill:** /r/ (made by rapid vibration of the tongue)
  - **Tap:** /ɾ/ (made by a quick touch of the tongue)
  - **Flap:** /ɾ/ (made by a quick movement of the tongue)
- ✚ **The Waveforms of Consonants:** Consonant waveforms show the acoustic characteristics of consonant sounds.
- ✚ **The Articulation of Vowel Sounds:** Vowels are made by the free flow of air through the vocal cords, with no obstruction by the articulators.
- ✚ **The Sounds of Vowels:** Vowels have different qualities (e.g., /i/, /e/, /a/) and can be long or short.
- ✚ **Long Vowels and Diphthongs**
  - **Long vowels:** prolonged vowel sounds (e.g., /i:/, /e:/)
  - **Diphthongs:** gliding vowel sounds (e.g., /ai/, /au/)
- ✚ **Introduction to Suprasegmentals:** Suprasegmentals refer to the prosodic features of speech, such as intonation, stress, and rhythm.

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- ✚ **Why Transcribe?** Transcription helps to represent speech sounds in a standardized way, facilitating communication and analysis.
- ✚ **Introduction to IPA:** The International Phonetic Alphabet (IPA) is a standardized system for transcribing speech sounds. The IPA chart displays the range of speech sounds, organized by place and manner of articulation.
- ✚ **Transcription of Vowels:** Vowels are transcribed using symbols that represent their quality and duration.
- ✚ **Transcription of Consonants:** Consonants are transcribed using symbols that represent their place, manner, and voicing.
- ✚ **Broad and Narrow Transcription**
  - **Broad transcription:** represents speech sounds in a general way, without detailed phonetic information
  - **Narrow transcription:** represents speech sounds in a detailed and precise way, including phonetic features
- ✚ **IPA Resource/ IPA Story:** The North Wind and the Sun is a famous fable used to demonstrate the International Phonetic Alphabet (IPA) transcription.
- ✚ **IPA Story:** Practice: Practicing transcribing the story using IPA symbols helps develop skills in phonetic transcription.
- ✚ **IPA Story: Broad Transcription:** Broad transcription represents the speech sounds in a general way, without detailed phonetic information.
- ✚ **Transcription Beyond Words:** Word stress indicates which syllable in a word receives the most emphasis.
- ✚ **Transcription Beyond Words: Word Stress:** Transcription goes beyond individual words to represent the sounds and rhythms of connected speech.
- ✚ **Transcription Beyond Words: Accent:** Accent refers to the unique pronunciation features of a particular region or language.
- ✚ **Transcription Beyond Words: Phrases:** Transcribing phrases shows how words are connected in speech, including rhythm and intonation.
- ✚ **Transcription Beyond Words: Rhythm and Beyond:** Rhythm and intonation are crucial aspects of speech, and transcription helps represent these features.
- ✚ **The Consonants of English:**
  - Stop consonants: /p/, /t/, /k/
  - Fricatives: /s/, /z/, /f/

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- Affricates: /tʃ/, /dʒ/
- Nasals: /m/, /n/, /ŋ/
- Approximants: /w/, /j/, /r/

✚ **Overlapping Gestures:** Consonant articulation often involves overlapping gestures, where multiple articulators are involved in producing a single sound.

✚ **Aspects of Connected Speech:** Connected speech involves co-articulation, where the articulation of one sound influences the production of adjacent sounds.

✚ **Co-articulation:** Co-articulation occurs when the articulation of one sound affects the production of adjacent sounds.

✚ **Rules for English Consonant Allophones (ECA):** ECA rules describe the variations in consonant pronunciation that occur in different contexts.

✚ **Rules for ECA:** Explanation: Understanding ECA rules helps predict how consonants will be pronounced in different words and contexts.

✚ **Introduction to Diacritics:** Diacritics are small marks added to IPA symbols to indicate detailed phonetic features such as aspiration and nasalization.

✚ **Diacritics and Detailed Transcription:** Diacritics allow for precise transcription of speech sounds, including subtle variations in pronunciation.

✚ **Aspiration:** Aspiration occurs when a consonant is pronounced with a burst of air, often represented by a superscript "h" in IPA transcription.

✚ **Nasalization:** Nasalization occurs when a sound is pronounced with air flowing through the nasal cavity, often represented by a tilde (~) in IPA transcription.

✚ **Velarization:** Velarization is a phonetic process where a sound is made with the tongue touching the velum (soft palate). Example: the "dark L" sound in "full" or "bell" (/fʊɫ/ or /bɛɫ/).

✚ **English Vowels:** English vowels are sounds made without blocking the airflow in the mouth. They can be monophthong (single, pure sounds) or diphthong (gliding sounds).

## ✚ **Vowel Quality**

Vowel quality refers to the distinct sound of a vowel. English vowels have different qualities, such as:

- /i/ (beat)

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- /e/ (bet)
- /a/ (bat)
- /o/ (bot)
- /u/ (but)

✚ **Auditory Vowel Space:** The auditory vowel space is a diagram showing the acoustic characteristics of vowels, helping to visualize their quality.

## ✚ **American and British Vowels**

American and British English have some differences in vowel pronunciation, such as:

- American /r/ is often pronounced with a more open and relaxed tongue position
- British /a/ in "bath" is often more open and longer than in American English

## ✚ **Diphthongs**

Diphthongs are gliding vowels, where the tongue moves from one position to another during pronunciation. Examples:

- /ai/ (buy)
- /au/ (how)
- /oi/ (boy)

✚ **Rhotic Vowels:** Rhotic vowels are vowels followed by the sound /r/. In some accents, like American English, /r/ is pronounced after vowels, while in others, like British English, it's often silent.

✚ **Unstressed Syllables:** Unstressed syllables have less emphasis and are often pronounced with a weaker, more central vowel sound. Example: the second syllable in "banana" (ba-NA-na).

## ✚ **Tense and Lax Vowels**

Tense vowels are pronounced with more tension and longer duration, while lax vowels are pronounced with less tension and shorter duration. Examples:

- Tense: /i/ (beat), /e/ (bet)
- Lax: /ɪ/ (bit), /ɛ/ (bet)

## ✚ **Fortis and Lenis Consonants**

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Fortis consonants are pronounced with more force and energy, while lenis consonants are pronounced with less force and energy. Examples:

- Fortis: /p/ (pat), /t/ (tot)
- Lenis: /b/ (bat), /d/ (dot)

✚ **Rules for English Vowel Allophones:** English vowel allophones are variations of vowel sounds that occur in different contexts. Examples:

- /i/ becomes [ɪ] before /ŋ/ (sing)
- /e/ becomes [ɛ] before /r/ (pet)

✚ **English Words and Sentences:** English words and sentences follow rules of phonology, syntax, and semantics to convey meaning.

✚ **Words in Connected Speech:** In connected speech, words are pronounced with influences from adjacent sounds, such as assimilation and elision.

## ✚ **Degree of stress**

Stress refers to the emphasis placed on a syllable or word in speech. English has three degrees of stress:

- Primary stress (strongest)
- Secondary stress
- Unstressed (weakest)

✚ **Sentence Rhythm:** Sentence rhythm refers to the pattern of stressed and unstressed syllables in speech.

✚ **Intonation:** Intonation refers to the rise and fall of pitch when speaking, used to convey meaning and attitude.

✚ **Target Tones:** Target tones are the pitch levels aimed for when speaking, which can be high, mid, or low.

## ✚ **Airstream Mechanisms**

Airstream mechanisms refer to the ways in which air is used to produce speech sounds:

- Pulmonic (using lungs)
- Glottalic (using vocal cords)
- Velaric (using velum)

## ✚ **Summary of the Airstream Mechanisms**

- Pulmonic: most common, uses lungs

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- Glottalic: used for glottal stops and some consonants
- Velaric: used for some consonants and vowels

✚ **PHONATION:** Phonation refers to the production of sound by the vocal cords.

## ✚ States of the Glottis

- Open (voicing)
- Closed (stop)
- Ajar (fricative)

## ✚ Phonation Types

- Voicing (vocal cords vibrate)
- Unvoicing (vocal cords don't vibrate)

✚ **Voicing and Consonants:** Voicing affects consonant pronunciation, with voiced consonants having vocal cord vibration and voiceless consonants having no vibration. Examples:

- Voiced: /b/ (bat), /d/ (dot)
- Voiceless: /p/ (pat), /t/ (tot)

✚ **Voice Onset Time (VOT):** Voice Onset Time (VOT) refers to the time it takes for the voice to start after the release of a consonant.

## ✚ VOT Explanation

VOT is important because it helps distinguish between similar sounds. For example, the difference between /p/ and /b/ lies in their VOT:

- /p/ has a long VOT (around 60-80 ms)
- /b/ has a short VOT (around 10-20 ms)

✚ **Importance of VOT:** VOT is crucial for accurate speech perception, as it helps listeners distinguish between similar sounds.

## ✚ Types of VOT

- Positive VOT (voice starts after consonant release)
- Negative VOT (voice starts before consonant release)

✚ **Consonantal Gestures:** Consonantal gestures refer to the movements and positions of the articulators (lips, tongue, etc.) that produce consonant sounds.

✚ **Articulatory Targets:** Articulatory targets are the specific places and manners of articulation that produce consonant sounds.

## ✚ Types of Articulatory Gestures



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- Stops (e.g., /p/, /t/)
- Nasals (e.g., /m/, /n/)
- Fricatives (e.g., /s/, /z/)

## Stops

Stops are consonants made by blocking the airflow with the tongue or lips, then releasing it. Examples:

- /p/ (pat)
- /t/ (tot)
- /k/ (cat)

## Nasals

Nasals are consonants made by allowing air to flow through the nasal cavity. Examples:

- /m/ (mat)
- /n/ (nat)
- /ŋ/ (sing)

## Fricatives

Fricatives are consonants made by directing air through a narrow channel, causing friction. Examples:

- /s/ (sat)
- /z/ (zat)
- /f/ (fat)

**Groups by Maha Malik**

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